

VAIKE-MAARJA, Estonia

WMO: 261410

Lat: 59.1414N

Lon: 26.2308E

Elev: 122

StdP: 99.87

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-21.1	-18.1	-23.5	0.5	-20.8	-20.4	0.6	-17.8	10.2	-0.7	9.5	-0.8	1.9	120	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	26.9	18.9	25.3	18.1	23.5	17.1	20.2	25.0	19.1	23.6	18.0	22.1	3.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	13.5	22.6	17.4	12.6	21.3	16.2	11.7	20.2	58.6	24.8	54.6	23.7	51.3	22.2	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.4	8.3	7.5	DB	-24.3	29.0	4.5	1.7	-27.5	30.2	-30.1	31.2	-32.6	32.2	-35.9	33.4	(4)
(5)			WB	-24.5	21.9	4.4	1.2	-27.6	22.8	-30.2	23.5	-32.7	24.1	-35.8	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-5.5	-5.6	-2.2	4.5	10.0	14.2	17.2	15.8	11.2	5.3	0.5	-2.9	(6)	
	DBStd	9.14	6.00	6.19	4.58	4.22	4.36	3.33	3.16	3.01	3.28	3.87	4.27	5.45	(7)	
	HDD10.0	2387	480	438	377	171	53	4	0	1	26	152	286	401	(8)	
	HDD18.3	4816	738	671	636	413	259	132	61	91	215	406	536	659	(9)	
	CDD10.0	658	0	0	0	7	54	129	222	179	61	5	0	0	(10)	
	CDD18.3	44	0	0	0	0	2	6	25	11	0	0	0	0	(11)	
	CDH23.3	402	0	0	0	1	22	62	216	100	2	0	0	0	(12)	
	CDH26.7	58	0	0	0	0	2	3	39	13	0	0	0	0	(13)	
(14) Wind	WSAvg	3.7	4.1	4.1	4.0	3.7	3.5	3.5	3.0	3.1	3.4	3.7	4.1	4.5	(14)	
(15) Precipitation	PrecAvg	681	48	38	35	32	47	77	78	88	59	71	59	50	(15)	
	PrecMax	901	93	93	63	63	112	144	154	187	105	127	106	99	(16)	
	PrecMin	441	10	7	11	9	8	14	8	5	16	9	11	25	(17)	
	PrecStd	105	19	20	16	15	26	36	36	50	27	24	22	18	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	4.6	11.4	20.9	26.0	27.1	29.7	28.2	23.0	15.0	9.3	7.9	(19)	
		MCWB	4.2	3.2	6.4	11.9	17.2	18.5	20.5	19.1	16.2	12.3	8.5	6.9	(20)	
	2%	DB	3.2	3.3	7.2	17.7	22.8	25.4	27.7	26.0	20.0	13.1	8.3	6.1	(21)	
		MCWB	2.7	2.4	3.9	10.6	15.5	17.5	19.7	18.3	15.2	11.6	7.6	5.4	(22)	
	5%	DB	2.1	2.4	4.8	14.7	20.6	23.0	25.8	24.0	18.1	11.9	7.2	4.7	(23)	
		MCWB	1.6	1.7	2.6	8.8	13.8	16.4	19.0	17.3	14.0	10.8	6.6	4.1	(24)	
	10%	DB	1.2	1.4	3.1	12.0	18.3	21.0	23.9	22.0	16.5	10.7	6.1	3.2	(25)	
		MCWB	0.8	0.9	1.5	7.3	12.1	15.2	18.0	16.6	13.5	9.6	5.6	2.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	3.7	6.7	13.3	18.8	20.2	22.0	21.2	17.1	13.0	8.7	7.1	(27)	
		MCDB	4.8	4.1	10.4	19.2	23.8	25.1	27.4	25.8	20.7	14.0	9.0	7.8	(28)	
	2%	WB	2.7	2.6	4.3	11.0	16.6	18.8	20.8	19.2	15.8	11.9	7.7	5.5	(29)	
		MCDB	3.0	3.2	6.5	16.3	21.1	23.1	25.8	23.8	18.7	12.7	8.2	6.0	(30)	
	5%	WB	1.7	1.7	2.9	9.4	14.8	17.3	19.7	18.2	14.9	10.9	6.7	4.1	(31)	
		MCDB	2.0	2.2	4.3	14.2	19.2	21.3	24.1	22.2	17.4	11.8	7.1	4.6	(32)	
	10%	WB	0.8	0.9	1.9	7.7	12.9	15.9	18.6	17.2	13.9	9.8	5.6	2.7	(33)	
		MCDB	1.2	1.4	2.9	11.3	17.1	20.0	23.0	21.1	15.8	10.6	6.1	3.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.1	6.6	8.8	10.4	9.4	9.4	9.5	7.8	5.5	3.7	3.9	(35)	
		MCDBR	3.4	4.2	8.4	13.8	13.4	12.7	12.2	12.7	10.2	6.5	4.2	3.6	(36)	
	5% WB	MCWBR	3.4	3.8	5.7	7.7	7.1	6.4	5.6	6.1	5.9	4.9	4.0	3.5	(37)	
		MCDWR	3.4	4.0	7.1	12.5	11.9	10.7	10.9	10.7	8.9	5.6	4.1	3.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.269	0.301	0.327	0.367	0.363	0.361	0.383	0.382	0.350	0.330	0.305	0.258	(40)	
		taud	2.111	2.138	2.189	2.300	2.391	2.426	2.382	2.384	2.463	2.408	2.205	2.080	(41)	
	Edn at Noon	Edn at Noon	520	698	800	835	867	874	841	813	784	678	475	392	(42)	
		Edh at Noon	48	78	101	108	106	104	107	99	78	61	45	34	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.28	0.88	2.34	3.86	5.36	5.53	5.37	4.18	2.58	1.10	0.31	0.17	(44)	
	RadStd	RadStd	0.05	0.16	0.31	0.44	0.57	0.44	0.50	0.42	0.31	0.15	0.05	0.03	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	-155	-162	N/A	N/A	N/A

Nomenclature: See separate page